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Real-time uncombined PPP using BDS-3 PPP-B2b products with different multi-frequency integrations and refined stochastic model

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Abstract

BDS-3 geostationary orbit (GEO) satellites broadcast PPP-B2b real-time precise products to support real-time precise point positioning (RT-PPP) without dependence on the ground network communication. Since all BDS-3 satellites can provide multi-frequency signals, we investigate the effect of different multi-frequency integrations and the refinement of multi-frequency stochastic model for PPP-B2b RT-PPP. Given that the retrieval of PPP-B2b real-time precise corrections relies on the B1C and B2b frequencies, this study first investigates the kinematic uncombined (UC) RT-PPP performance of the B1C/B2b integration with a comparison to the conventional B1I/B3I integration. The results indicate that the convergence time (with a convergence threshold of 20 cm) of the RT-PPP with the B1C/B2b integration is shortened by 16 %, 15 %, and 12 % over the B1I/B3I integration in the east, north, and up directions, respectively. Further, this study compares the kinematic UC RT-PPP performance of the B1C/B2b dual-frequency integration, B1C/B2b/B3I triple-frequency integration, and B1C/B2b/B3I/B1I/B2a five-frequency integration. Compared with the dual-frequency integration, the positioning performance of the triple-frequency integration is slightly improved. By contrast, the performance improvement of the five-frequency integration is more significant, which can reach 6 %, 32 %, and 9 % for the convergence time, and 11 %, 9 %, and 6 % for the positioning accuracy in three directions, respectively. To further enhance the multi-frequency PPP performance, a refined stochastic model for the BDS-3 five-frequency integrated PPP-B2b RT-PPP is proposed by taking the inconsistency of signal quality among multi-frequency signals into account. After applying the refined stochastic model, the convergence time of the five-frequency kinematic RT-PPP is shortened by 10 %, 14 %, and 9 % to 32, 7, and 19 min in east, north, and up directions, respectively. The positioning accuracy (the root mean square of all the positioning errors excluding the first two hours) of the five-frequency RT-PPP with the refined stochastic model can reach the optimal level, which is 9.9, 6.5, and 14.0 cm in three directions, respectively.
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Keywords: BDS-3; Real-time; Precise point positioning; PPP-B2b; Multi-frequency integration; Stochastic model

1. Introduction

Precise point positioning (PPP) is a technique to utilize precise orbit and clock products to achieve static centimeter-to-millimeter and kinematic decimeter-to-centimeter positioning accuracy with a single Global Navigation Satellite System (GNSS) receiver (Malys and Jensen, 1990; Zumberge et al., 1997; Kouba and Héroux, 2001). Due to its high accuracy, the PPP technique has been extensively applied to Earth's crust deformation monitoring, near-real-time GPS meteorology, low-orbit satellite orbit determination, and precise positioning and navigation (Zhu et al., 2004; Calais et al., 2006). In the real-time PPP (RT-PPP), the precise orbits and clock corrections can be obtained via the Internet or via the satellite

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link. For the former, the precise products are generally transmitted to users using the Networked Transport of Radio Technical Commission for Maritime Services (RTCM) via Internet Protocol (NTRIP), such as the Real-Time Service (RTS) of the International GNSS Service (IGS). Currently, the orbital correction accuracy can reach a centimeter level, while the clock correction accuracy is better than 0.2 ns (Li et al., 2022). However, the potential Internet connectivity disruptions and network coverage limitations in remote areas severely hamper the applications of the Internet-based RT-PPP.

To facilitate the RT-PPP application, several constellations, such as Galileo, Quasi-Zenith Satellite System (QZSS), and BDS-3, have realized the broadcast of precise products via the satellite link. Specifically, Galileo started the free global High Accuracy Service (HAS) on the E6-B signal on January 24, 2023, with a horizontal accuracy of 20 cm and a vertical accuracy of 40 cm for the RT-PPP after a convergence of 100 s in European coverage area (European Union, 2022; EUSPA, 2020; European GNSS Service Center, 2023). Cabinet Office in Japan developed a new augmentation service, namely multi-GNSS advanced orbit and clock augmentation-precise point positioning (MADOCA-PPP), on September 30, 2022 (Cabinet Office, 2022), which realizes the RT-PPP in Asia-Oceania region by providing GNSS augmentation data via QZSS satellites. BDS-3 offered a regional PPP service, known as PPP-B2b to users in and around China. Specifically, BDS-3 broadcasts PPP-B2b signals containing precise satellite orbit and clock corrections information to users through the geostationary orbit (GEO) satellites, enabling RT-PPP free of charge without reliance on the Internet.

Since the release of the interface control document for the PPP-B2b spatial signals, numerous studies have been made to evaluate the performance of PPP-B2b products and the RT-PPP. Liu (2020) conducted an in-depth analysis of the PPP-B2b signal, optimizing the ACE-BOC multipopulational method to capture its signal characteristics. Experimental results revealed horizontal and vertical positioning accuracies of 0.15 and 0.3 m, respectively, with an average convergence time of less than 14 min based on BDS. Tang et al. (2022) conducted a comprehensive analysis of satellite orbit, clock, and Differential Code Bias (DCB) corrections contained in PPP-B2b information. They evaluated the accuracy of the RT-PPP based on PPP-B2b, demonstrating that the simulated kinematic positioning accuracy using BDS-3/GPS dual-system is better than 2.5 cm in the north direction, 3.5 cm in the east direction and 8.5 cm in the up direction after convergence. Liu et al. (2022) analyzed the positioning accuracy of PPP-B2b using data from IGS stations in and around China. The results indicated that the PPP-B2b-based PPP can achieve static centimeter-level and kinematic sub-meter-level positioning accuracy in China. However, in the boundary area of the PPP-B2b service, the positioning performance is significantly affected by the number of visible satellites. Wu et al. (2023) conducted comprehensive evaluations of RT-PPP using the PPP-B2b products on kinematic and highly kinematic platforms, and results indicated that a convergence time of 28.5 and 12.9 min could be achieved with BDS-3 only and BDS-3/GPS dual-system, respectively. In addition, Xu et al. (2023) estimated the spatial ranging error of each satellite for improving the positioning accuracy and reliability of PPP-B2b, and the ship-borne experiment results showed that the positioning accuracies in horizontal and vertical directions were improved by 36 % and 19 %, respectively.

Actually, a suitable stochastic model is important for parameter estimation in PPP processing (Koch, 1999). The main stochastic models for PPP include the a posterior model and the a priori model. The former mainly uses the a posterior residuals to iteratively correct the variances of observations at an epoch (Li et al., 2011; Parvazi et al., 2020), which can be well used to obtain the observation weights among different constellations, but the iterative processing usually costs too much computation time to suit for RT-PPP. In contrast, the a priori model utilizes some a priori indicators to determine the variance–covariance matrix of the observations, including the elevation-dependent and carrier-to-noise ratio (C/N_0) stochastic models (Gardan, 1995; Brunner et al., 1999). Recently, many scholars have conducted research and expansion on the two a priori stochastic models (Liu et al., 2024; Yuan et al., 2023). However, the weights of observations on different frequencies were mostly considered equal, and they did not consider the inconsistency of signal quality among multi-frequency signals.

In the existing research on PPP-B2b, most scholars selected the B1I/B3I observation integration by default in their RT-PPP implementation. It should be noted that the open service of BDS-3 refers to the B3I broadcast ephemeris, but the calculation of PPP-B2b corrections broadcast through the B2b channel must rely on the B1C broadcast ephemeris. Thus, for the dual-frequency users, it is preferable to use the B1C/B2b integration to realize PPP-B2b-based RT-PPP. In this study, the PPP-B2b-based RT-PPP performance is evaluated using the B1C/B2b integration with a comparison to the B1I/B3I integration. In addition, the multi-frequency uncombined (UC) PPP performance based on PPP-B2b is analyzed using different multi-frequency integrations. Because of the inconsistency of signal quality among multi-frequency signals, the observations of different frequencies cannot be simply regarded as equal weights. Therefore, this study further proposes a refined stochastic model on the basis of the traditional elevation-dependent stochastic model to improve the positioning performance of the multi-frequency RT-PPP.

2. PPP-B2b real-time precise corrections

The B2b signal broadcast by BDS-3 GEO satellites modulates the precise orbit and clock corrections for BDS-3/GPS satellites to enable RT-PPP. The correction information primarily encompasses pseudo-random number (PRN), satellite orbit, satellite clock, and DCB correction information. The specific correction method is provided in the following sections (CSNO, 2019b).

2.1. Satellite orbit corrections

The information type 2 included in the PPP-B2b information are the satellite orbit corrections and the user ranging accuracy index (URAI). The orbit correction information refers to the satellite position correction vector in the radial, tangential and normal directions. The satellite position vector is first obtained using the B1C broadcast ephemeris, and then the satellite orbit corrections are used to correct the position vector. The correction formula is provided below.

$$X_{pre\_orbit} = X_{broadcast} - \delta_X \tag{1}$$

where $X_{broadcast}$ is the satellite position calculated by the B1C broadcast ephemeris, X_{pre_orbit} is the satellite position after corrections, and δ_X is the orbital correction vector calculated by Eq. (2).

$$\begin{cases} e_{radial} = \frac{r}{|r|} \\ e_{cross} = \frac{r \times \dot{r}}{|r \times \dot{r}|} \\ e_{along} = e_{cross} \times e_{radial} \\ \delta_X = \begin{bmatrix} e_{radial} & e_{along} & e_{cross} \end{bmatrix} \cdot \delta_O \end{cases} \tag{2}$$

where δ_O is the satellite orbit correction vector included in the PPP-B2b information, r and $\dot{r}$ are the position and velocity vectors of the satellites calculated by broadcast ephemeris, e_{radial}, e_{along} and e_{cross} are the unit vectors in the radial, tangential and normal directions, respectively.

However, The PPP-B2b products provide the antenna phase center (APC) satellite position of the B3I signal for BDS-3, which is different from the center-of-mass satellite position provided by IGS products (Tao et al., 2021). Therefore, we use the ANTenna Exchange format (ANTEX) file provided by IGS to correct the reference of the satellite position from B3I signal to the phase center of each frequency during the UC PPP processing.

2.2. Satellite clock corrections

The information type 4 included in the PPP-B2b denotes the satellite clock corrections. The clock corrections of each satellite include *IOD Corr* and C_0 two parts. *IOD Corr* is the version number of the correction information, which is used to match the version number of the information type 2 (satellite orbit corrections). C_0 is the clock correction, which is valid in the range of ±26.2128m. The correction formula for satellite clock errors is provided in Eq. (3).

$$t_{pre\_sat\_clk} = t_{broadcast} - \frac{C_0}{c} \tag{3}$$

where, $t_{broadcast}$ is the satellite clock calculated by the B1C broadcast ephemeris, $t_{pre\_sat\_clk}$ is the satellite clock after correction, and c is the speed of light.

Similarly, the PPP-B2b clock corrections refer to B3I signal (Tao et al., 2021), which can't directly be used in PPP processing. Therefore, we use the DCB corrections provided by the PPP-B2b products to correct the code observations on each frequency, so that all of them only contain the satellite code hardware delay of B3I signal.

2.3. DCB corrections

The information type 3 contained in the PPP-B2b information is the DCB correction. Each set of information encompasses the count of satellites and the respective count of DCB for each satellite. As these values are variable, the DCB correction algorithm involves matching the PRN through the issue of data PRN (IODP) which comes from the PPP-B2b products. The IODP defines the version of satellite mask and simultaneously presents in PNR mask, orbit correction, and DCB correction information. The specific correction formula for DCB is provided in Eq. (4).

$$\tilde{P}_i = P_i - DCB_i \tag{4}$$

where, $\tilde{P}_i$ is the corrected code observations on frequency i, P_i is the original code observation, and DCB_i is the DCB correction on frequency i. Specifically, the DCB corrections provided by the PPP-B2b products includes DCB (B1I), DCB (B1C), DCB (B2a) and DCB (B2b). Both the BDS-3 broadcast clock parameters and the PPP-B2b precise clock offsets refer to the B3I signal. The DCBs mentioned above refer to differential code biases between the corresponding signal and the B3I signal, such as DCB (B1I) refers to the differential code bias between B1I and B3I. Therefore, we directly subtract the DCB corrections in Eq. (4) to maintain the consistent usage of satellite clock offsets.

3. BDS-3 multi-frequency UC PPP model

3.1. Observation model

Differentiating in the treatment of ionospheric delay, the commonly employed PPP observation models primarily include the ionospheric-free (IF) combination model and the UC model. The UC model estimates the ionospheric delay parameters in the original observation equation, and thus it brings more parameters to be estimated. The IF combination uses the inter-frequency combined observations to eliminate the first-order ionospheric delay errors, but it amplifies the measurement noise. Although the two models are different, their positioning performance is very similar in the dual-frequency RT-PPP (Pan et al., 2021). It should be noted that both the IF and UC can allow processing more than two frequencies. However, in multi-frequency case, the observation model of the IF becomes more diverse, while the UC one exhibits better extendibility and flexibility. To facilitate multi-frequency positioning, this study chooses the UC model for data processing, and its observation equations are as follows (Pan et al., 2019),

$$P_{r,i}^{s} = \rho_r^s + cdt_r - cdt^s + T_r^s + \gamma_i I_i^s + d_{r,i} + d_i^s + \varepsilon_{P,i} \tag{5}$$

$$L_{r,i}^{s} = \rho_r^s + cdt_r - cdt^s + T_r^s - \gamma_i I_i^s + N_i + b_{r,i} + b_i^s + \varepsilon_{L,i} \tag{6}$$

where $P_{r,i}^s$ and $L_{r,i}^s$ are the receiver code and carrier phase observations in meters respectively, and ρ_r^s ($\rho = \sqrt{(x_r - x^s)^2 + (y_r - y^s)^2 + (z_r - z^s)^2}$) is the geometric range from the satellite to the receiver, (x_r, y_r, z_r) and (x^s, y^s, z^s) represent the 3D coordinates of the receiver and satellite in the Earth-Centered Earth-Fixed (ECEF) coordinate system, respectively, cdt_r and cdt^s are the clock errors of the receiver and satellite, respectively, T_r^s represents the slant tropospheric delay, $\gamma_i = \lambda_i^2/\lambda_1^2$ represents the ionospheric delay factor, where λ is the carrier wavelength, I_i^s is the first-order slant ionospheric delay on the first frequency (B1I or B1C), N represents the phase ambiguity, $d_{r,i}$ and d_i^s are the code hardware delays of the receiver and the satellite, $b_{r,i}$ and b_i^s are the phase hardware delays of the receiver and the satellite, and ε represents the measurement noises.

After applying the PPP-B2b corrections and the other error corrections that need to be considered in PPP, the BDS-3 multi-frequency UC PPP observation model can be expressed as,

$$\begin{cases} p_1 = \mu \cdot X + \overline{cdt}_{r,UC} + \bar{I}_1 \cdot \gamma_1 + m \cdot T_r \\ p_2 = \mu \cdot X + \overline{cdt}_{r,UC} + \bar{I}_1 \cdot \gamma_2 + m \cdot T_r \\ p_i = \mu \cdot X + \overline{cdt}_{r,UC} + \bar{I}_1 \cdot \gamma_i + m \cdot T_r + IFB_{UC,i} \\ l_1 = \mu \cdot X + \overline{cdt}_{r,UC} - \bar{I}_1 \cdot \gamma_1 + m \cdot T_r + \bar{N}_1 \\ l_2 = \mu \cdot X + \overline{cdt}_{r,UC} - \bar{I}_1 \cdot \gamma_2 + m \cdot T_r + \bar{N}_2 \\ l_i = \mu \cdot X + \overline{cdt}_{r,UC} - \bar{I}_1 \cdot \gamma_i + m \cdot T_r + \bar{N}_i \end{cases} \tag{7}$$

where p and l represent the observation-minus-computed (O-C) code and phase observables in meters, respectively, μ is a unit vector in the line-of-sight direction from the receiver to the satellite, X is the 3D coordinates of the receiver (station), and $\overline{cdt}_{r,UC}$ is the receiver clock offset estimated in the UC model which absorbs the receiver code hardware delays. $\bar{I}_1$ is the estimated first-order slant ionospheric delay on the first frequency which absorbs the receiver code hardware delays, T_r is the zenith tropospheric wet delay (ZWD), m represents the wet delay projection function, and $IFB_{UC,i}$ $(i \geq 3)$ is the inter-frequency bias (IFB) on the *i-th* frequency, which only appears in the code observation equation of the third and higher frequencies, $\bar{N}_i$ represents the estimated ambiguity parameters for each frequency which absorbs the code and phase hardware delays of the receiver and satellite and loses its integer characteristics, so it can not directly allow integer ambiguity fixing. If need to fix the ambiguity, the PPP ambiguity resolution (AR) model should be established (Ge et al., 2008). The parameters to be estimated in the BDS-3 multi-frequency UC PPP model include receiver coordinates, receiver clock offsets, ionospheric delay, ZWD, IFB and phase ambiguities of each frequency. If there are n satellites observed at an epoch, then the parameters to be estimated at that epoch can be expressed as,

$$\boldsymbol{X}_{UC} = \begin{bmatrix} X & \overline{cdt}_{r,UC} & \bar{I}_1 & T_r & IFB_{UC,3} \cdots IFB_{UC,i} & \bar{N}_1^1 \cdots \bar{N}_1^n & \cdots & \bar{N}_i^1 \cdots \bar{N}_i^n \end{bmatrix} \tag{8}$$

It should be noted that the observation model established here is only applicable to BDS single-system case, and some modifications need to be made when it is applied to multi-GNSS combination. The receiver clock offset, ionospheric delay, IFB, and ambiguity parameters should be extended to incorporate multi-GNSS observations, while the 3D coordinates of the receiver and the ZWD parameters should be maintained.

3.2. Widely used stochastic model

The commonly used stochastic models in PPP mainly include the equal weight model, elevation stochastic model and C/N_0 stochastic model. The quality of the stochastic models directly affects the positioning accuracy. Considering that GNSS receivers suffer from the effect of various errors in the process of receiving satellite signals, and most of these errors are related to the elevation angles of satellites, this study chooses the elevation-dependent model as shown below (Gardan, 1995),

$$\sigma_i^s = \sqrt{a^2 + \frac{b^2}{sin^2(E_i^s)}} \tag{9}$$

where, a and b are the empirical values, generally $a = 4mm, b = 3mm$, and E is the satellite elevation angle.

With the completion of the BDS-3 construction, the frequency number of the BDS observations has increased to seven, i.e. B1I, B2I, B3I, B1C, B2b, B2a, and B2a + b and the quality of observations at different frequencies is different. For example, the signal quality of B1C and B1I is worse than that of B2a, B3I and B2b (Zhang et al. 2017). However, the traditional elevation-dependent stochastic model does not distinguish this. In this study, a refined elevation stochastic model is proposed by taking the signal quality at different frequencies into account.

4. Refinement of elevation-dependent stochastic model

In multi-frequency PPP processing, the observations of different frequencies are usually treated as equal weights without taking the inconsistency of signal quality among multi-frequency signals into account (Xu et al., 2022). This study optimizes the traditional elevation-dependent stochastic model from the perspective of signal quality.

First, the noise and multipath combinations (MP) for each frequency of BDS-3 code observations are acquired. Subsequently, the carrier phase quartic difference is made to extract the noise for each frequency's carrier phase observations. Finally, we use the elevation-dependent stochastic model to determine the initial weights of each observation and then utilize the code MP values and the phase noises to modify the elevation-dependent stochastic model.

4.1. Assessment of code observation noises

MP is obtained from the combination of single-frequency code observations and dual-frequency carrier phase observations, specifically expressed as (Estey and Meertens, 1999),

$$\begin{cases} MP_i = P_i - \frac{f_i^2+f_j^2}{f_i^2-f_j^2}L_i + \frac{2f_j^2}{f_i^2-f_j^2}L_j \\ = M_i - \frac{f_i^2+f_j^2}{f_i^2-f_j^2}m_i + \frac{2f_j^2}{f_i^2-f_j^2}m_j + N + \varepsilon_i \end{cases} \tag{10}$$

where, i and j ($i \neq j$) denote two different frequencies, which cannot be too close, MP_i is code MP value, f is the signal frequency, M is the multipath error of the code observations, and m is the multipath effect of the carrier phase observations, which has little effect on the MP value due to its small size and is usually ignored, N contains the phase ambiguity and the signal delay information, which can be regarded as a constant in the case of continuous observations free of cycle slips.

From Eq. (10), it is evident that the MP eliminates the influence of geometric range from satellite to receiver, clock errors, and atmospheric delay errors. It should be noted that the MP values are related to code multipath, which is site-related. However, the IGS stations are usually in open environments, where the code multipath effects are much limited. Therefore, the MP values can be used as an important index to measure the quality of code observations. The MP values primarily include residual phase ambiguity, multipath errors, and code noises. When the ambiguity is removed as a constant, the MP values are only related to the code multipath and the noise (Estey and Meertens, 1999), and then the MP values can be used to measure the quality of the code observations.

4.2. Assessment of phase observation noises

The carrier phase quartic difference method (Li and Huang, 2010) is usually used to calculate the noises of carrier phase observations. In this method, inter-satellite single-difference which can eliminates some receiver related errors, including the receiver clock errors and the receiver hardware delays, and inter-epoch triple-difference are made using carrier phase observations without cycle slips. The formula is as follows.

$$\Delta l^{u,v}(t) = \Delta\rho^{u,v} + \Delta cdt^{s,u,v} + \Delta N^{u,v} + \Delta I^{u,v} + \Delta T^{u,v} + \Delta\varepsilon_l^{u,v} \tag{11}$$

where, Δl represents the carrier phase after inter-satellite single difference in meters, u and v represent non-reference satellite and reference satellite, respectively. ΔI and ΔT are the ionospheric delay and tropospheric delay after the inter-satellite single difference, respectively. Eq. (11) is obtained from the carrier phase inter-satellite single difference, eliminating the receiver clock's influence on the carrier phase. Then the inter-epoch triple difference is further made on the phase residuals to eliminate the influence of the other residuals on the carrier phase. The left residual values only include the phase noise. The quartic difference of carrier phase can be expressed as:

$$\Delta L = \Delta l^{u,v}(t) - 3\Delta l^{u,v}(t+1) + 3\Delta l^{u,v}(t+2) - \Delta l^{u,v}(t+3) \tag{12}$$

where, ΔL is the quartic difference of the carrier phase observations in meters.

4.3. Modification of elevation-dependent stochastic model

After obtaining the code MP values and carrier phase noises, they are used to modify the elevation-dependent stochastic model to assign different weights to the code and phase observations of different frequencies by taking the B1I frequency as a reference frequency. The modified stochastic model is expressed as:

$$\sigma_{P_i} = \frac{MP_{B1I}}{MP_i}\sigma_{P_i}^e \tag{13}$$

$$\sigma_{L_i} = \frac{\Delta L_{B1I}}{\Delta L_i}\sigma_{L_i}^e \tag{14}$$

where, $\sigma_{P_i}^e$ and $\sigma_{L_i}^e$ are the weights of the code and phase observations of each frequency in the elevation-dependent stochastic model (by Eq. (9)), MP_{B1I} and MP_i are the MP values of the code observations at B1I and the other frequencies, ΔL_{B1I} and ΔL_i are the noises of the phase observations at B1I and the other frequencies, σ_{P_i} and σ_{L_i} are the refined variances of code and phase observations of each frequency. Specially, we use the elevation-dependent stochastic model and the numerical values (a = 4 mm

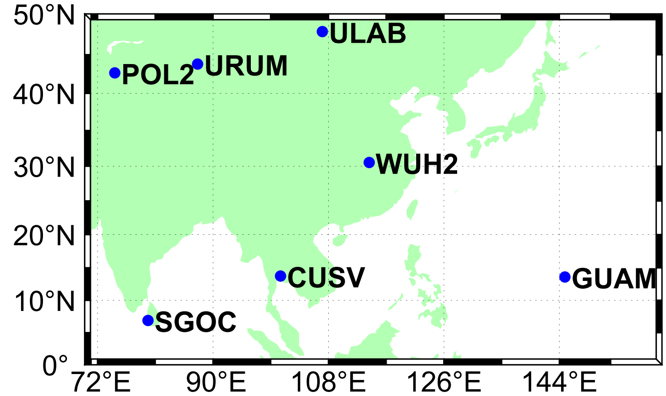

Fig. 1. Distribution of stations.

and b = 3 mm) to determine the variance of the observations of B1I. Although these numerical values may not fully reflect the true noise level of B1I, the marginal differences in observation variances caused by them have little impact on the accuracy of RT-PPP. In contrast, the ratios of observation variances among different frequencies are more important.

5. Results and discussion

5.1. Data sets and processing strategies

Observation data spanning days of year (DOY) 303–309 in 2022 were chosen at seven IGS stations in the Asia-Pacific region. Their geographical distribution is shown in Fig. 1. The observation data and BDS-3 PPP-B2b information received by the station at Central South University, China, were employed to realize simulated kinematic (static-kinematic) UC RT-PPP, and it is noted that all the results are calculated in post-processing using RT settings.

Table 1 presents the GNSS equipments (receiver and antenna) of the seven stations above. From the table, the SGOC, ULAB, URUM, and WUH2 stations have the same GNSS equipment, while the remaining three stations correspond to different GNSS equipments.

The RT-PPP processing strategies are shown in Table 2. For the weight determination strategy, the traditional elevation angle stochastic model is used in sections 5.2 and 5.3, while the refined stochastic model is adopted in section 5.4.

Table 1
GNSS equipment (receiver and antenna) for each station.

Stations	Receiver	Antenna
CUSV	JAVAD TRE_3 DELTA	JAVRINGANT_DM NONE
GUAM	SEPT POLARX5	JAVRINGANT_DM SCIS
POL2	JAVAD TRE_3 DELTA	TPSCR.G3 NONE
SGOC	JAVAD TRE_3	JAVRINGANT_G5T NONE
ULAB	JAVAD TRE_3	JAVRINGANT_G5T NONE
URUM	JAVAD TRE_3	JAVRINGANT_G5T NONE
WUH2	JAVAD TRE_3	JAVRINGANT_G5T NONE

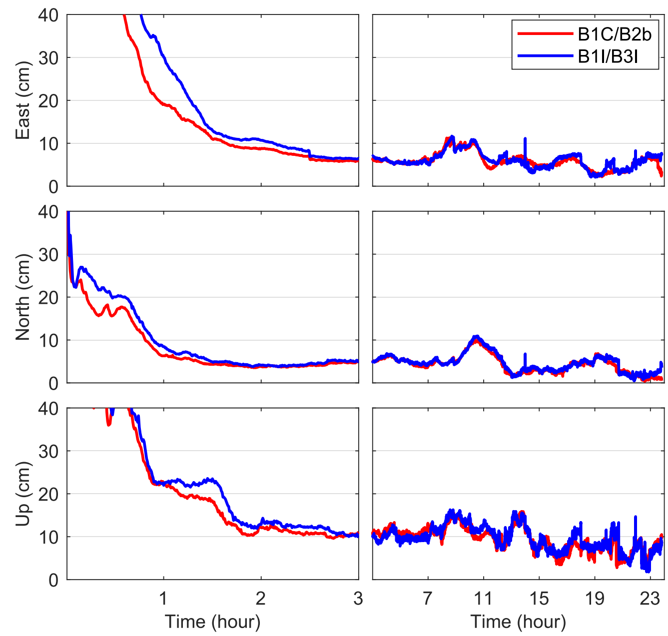

Fig. 2. Epoch-wise RMS statistics of kinematic RT-PPP positioning errors of B1I/B3I and B1C/B2b integrations using the datasets from 7 stations on DOY 303–309, 2022 (the left panel displays the positioning errors in the first three hours at a converging stage, and the right panel displays those after a complete convergence).

Table 2
Processing strategies for RT-PPP based on PPP-B2b.

Items	RT-PPP Processing strategies
Period of data	DOY 303–309, 2022
Observation interval	30 s
Satellite system	BDS-3 (excluding GEO satellites)
Elevation cut-off	10°
Weight strategy	elevation angle stochastic model or refined stochastic model
Estimation	Kalman filter
Satellite orbit	B1C Broadcast ephemeris + PPP-B2b orbit corrections
Satellite clock	B1C broadcast ephemeris + PPP-B2b clock corrections
Satellite DCB	PPP-B2b DCB corrections
Wind-Up	Model correction
Relativistic effects	Model correction
Tides	Corrected using IERS Convention 2010, including polar tides, ocean loading, and Solid Earth tides
PCO and PCV	Corrected using the ANTEX file provided by IGS
Tropospheric delay	Saastamoinen model for dry delayMapping slant delays to zenith delays with global mapping function (GMF), and estimated as random-walk (RW) variables for wet delay
Ionospheric delay	Estimated as RW
Receiver coordinates	Estimated as white noise
Receiver clock	Estimated as white noise
IFB	Estimated as RW
Phase ambiguity	Estimated as float constants

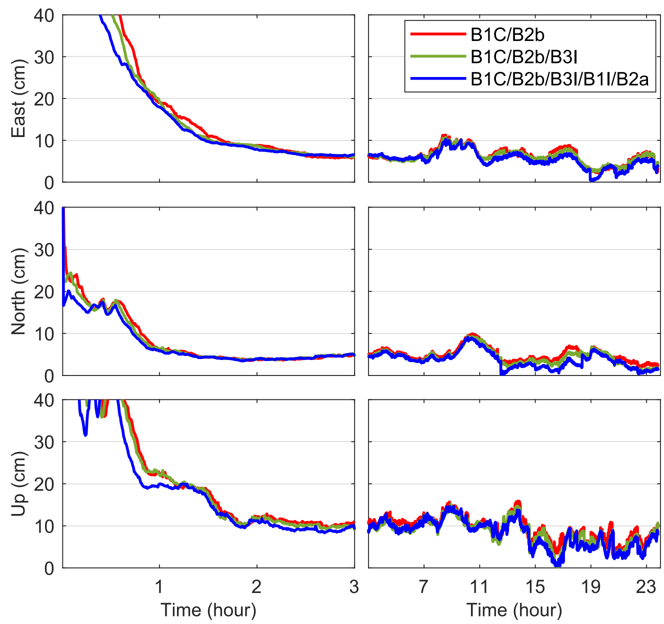

Fig. 3. Epoch-wise RMS statistics of multi-frequency kinematic RT-PPP positioning errors using the datasets from 7 stations on DOY 303–309, 2022 (the left panel displays the positioning errors in the first three hours at a converging stage, and the right panel displays those after a complete convergence).

Table 3
Dual-frequency kinematic RT-PPP performance using B1I/B3I and B1C/B2b integrations.

Direction	Convergence time (minute)		Positioning accuracy (cm)	
	B1I/B3I	B1C/B2b	B1I/B3I	B1C/B2b
East	44.8	37.7	11.6	11.5
North	13.4	11.4	7.3	6.8
Up	26.1	23.0	15.4	15.2

Table 4
Multi-frequency kinematic RT-PPP performance.

Direction	Convergence time (minute)			Positioning accuracy (cm)		
	B1C/B2b	B1C/B2b/B3I	B1C/B2b/B3I/B1I/B2a	B1C/B2b	B1C/B2b/B3I	B1C/B2b/B3I/B1I/B2a
East	37.7	37.1	35.5	11.5	11.2	10.2
North	11.4	11.5	7.8	6.8	6.8	6.2
Up	23.0	22.5	21.0	15.2	15.1	14.3

The station coordinates provided by the IGS SINEX file are used as the coordinate reference to calculate the positioning errors (site-centric coordinates based on the reference coordinates), and then the convergence time is calculated following a criterion that the positioning errors of the three directions (East, North, and Up) at 10 consecutive epochs are less than or equal to 20 cm. Finally, the RMS (root mean square) of positioning errors of the last 22 h is calculated. That is, the errors of the first two hours

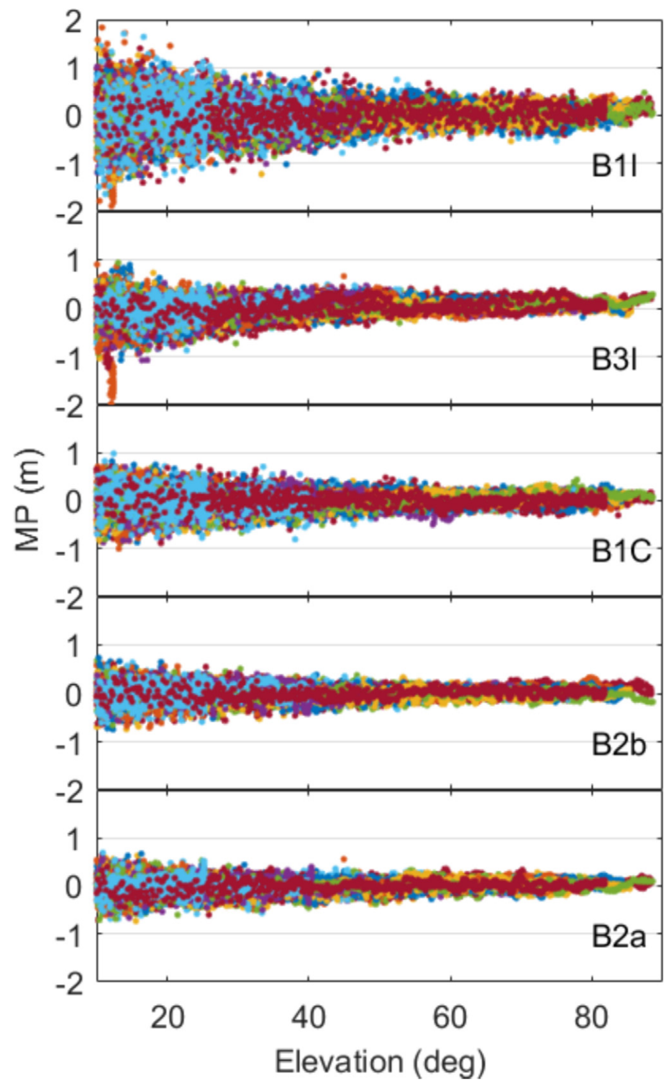

Fig. 4. Scatter chart of code MP values against elevation angles for each frequency at URUM station on DOY 303, 2022.

are excluded for each 24-h dataset due to avoid the impact of excessive positioning errors during the convergence process. Then, the performance of RT-PPP is evaluated in terms of the convergence time and the positioning RMS errors in three coordinate directions. To clearly see the difference in convergence speed of UC RT-PPP under different strategies, we divided the error curves (Figs. 2, 3, and 6) into two parts with the GPS time 3:00 as the boundary.

5.2. Dual-frequency kinematic PPP-B2b RT-PPP

The B1C/B2b dual-frequency integration is selected for kinematic RT-PPP and its positioning performance is compared with the traditional B1I/B3I dual-frequency integration. Fig. 2 shows the epoch-wise RMS statistics of RT-PPP positioning errors of B1I/B3I and B1C/B2b integrations using the datasets from 7 stations on DOY 303–309, 2022. The positioning errors at the specific epoch with the same time spans from the first epoch in the respective

Table 5
RMS statistical results of code MP for each frequency at each station.

Frequency	Code MP (m)							Mean (m)	Ratio to B1I
	CUSV	GUAM	POL2	SGOC	ULAB	URUM	WUH2		
B1I	0.18	0.24	0.22	0.25	0.22	0.19	0.21	0.216	1.00
B3I	0.07	0.14	0.11	0.20	0.15	0.13	0.15	0.136	0.63
B1C	0.11	0.17	0.12	0.20	0.13	0.10	0.14	0.139	0.64
B2b	0.06	0.13	0.09	0.21	0.12	0.08	0.11	0.114	0.53
B2a	0.07	0.13	0.09	0.14	0.11	0.08	0.13	0.107	0.50

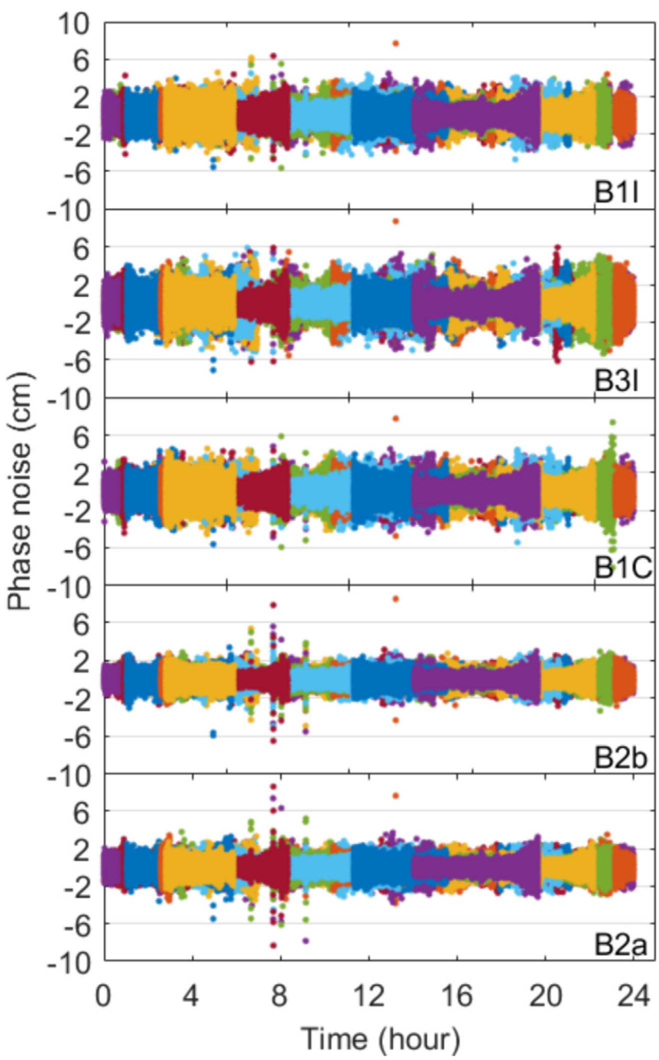

Fig. 5. Scatter chart of phase noises for each frequency at URUM station on DOY 303, 2022.

sessions for all 24-hour sessions are used to compute the epoch-wise RMS statistics. In this figure, the B1C/B2b dual-frequency integration performs better than the traditional B1I/B3I integration in both the convergence time and positioning accuracy. The reason for the step changes of epoch-wise RMS statistics at 2:30:00 in the east direction is that the observations are absent after this at a station.

The RMS errors in three directions along with the convergence time at all stations during DOY 303–309, 2022, were computed, and their mean values are displayed in Table 3. According to the statistical results, the kinematic RT-PPP with the B1C/B2b integration performs better when compared with the B1I/B3I dual-frequency integration. The convergence time is shortened by 16 %, 15 %, and 12 % in the east, north and up directions, respectively. The positioning accuracy has a marginal improvement in the three directions.

Based on the statistical results, it is evident that the RT-PPP performance using the B1C/B2b integration surpasses that of the B1I/B3I integration. For the convergence time of dual-frequency PPP, the B1C/B2b integration has a faster convergence speed without compromising the positioning accuracy than the B1I/B3I integration. Therefore, we conclude that the RT-PPP performance using the B1C/B2b integration surpasses that of the B1I/B3I integration. The reason for this result may be that B1C and B2b, as the new generation signals of BDS-3, have better anti-interference performance and higher accuracy (He, 2020). This conclusion offers a novel frequency integration option for future RT-PPP implementations based on the PPP-B2b information. Furthermore, given that B1C and B2b are two essential frequencies for the retrieval of PPP-B2b real-time precise corrections, dual-frequency receivers can opt for the B1C/B2b integration directly in the RT-PPP without the need to configure additional channels for receiving B1I and B3I signals. This not only help to improve the RT-PPP performance but also contributes to cost savings in receiver production.

5.3. Multi-frequency kinematic PPP-B2b RT-PPP

With the development of BDS-3, the number of frequencies of BDS-3 has increased to six, and the multi-frequency PPP has become a main trend in future development. To assess the positioning performance of the multi-frequency kinematic RT-PPP based on the B1C/B2b integration, this paper analyzes the RT-PPP performance based on PPP-B2b using different multi-frequency integrations. Fig. 3 shows the epoch-wise RMS statistics of kinematic RT-PPP positioning errors using the datasets from 7 stations on DOY 303–309, 2022, using three different frequency

Table 6
Statistical results of phase noises for each frequency at each station.

Frequency	Carrier phase noise (mm)							Mean (mm)	Ratio to B1I
	CUSV	GUAM	POL2	SGOC	ULAB	URUM	WUH2		
B1I	7.00	5.20	6.30	7.30	6.50	6.70	7.00	6.538	1.00
B3I	6.50	4.80	9.30	8.80	8.00	7.60	8.00	7.488	1.15
B1C	7.60	5.90	6.60	7.60	7.00	7.10	7.60	7.000	1.07
B2b	5.40	5.70	7.00	5.30	4.70	4.60	4.90	5.588	0.85
B2a	6.10	5.60	6.40	6.70	5.00	5.10	5.60	6.050	0.93

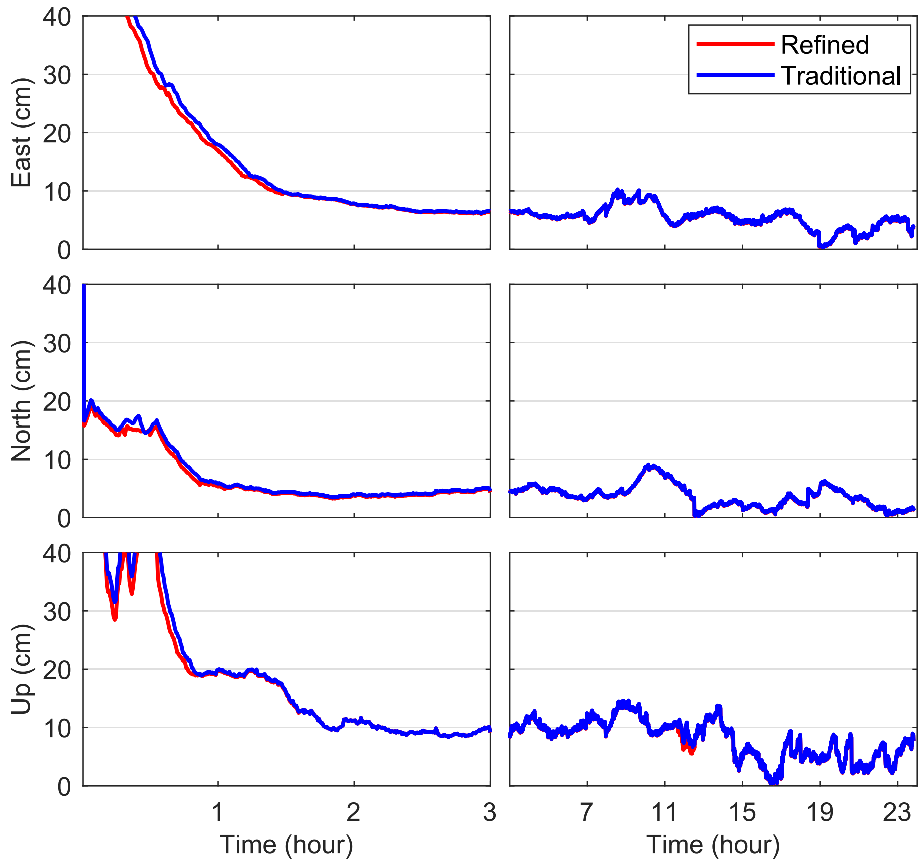

Fig. 6. Epoch-wise RMS statistics of RT-PPP positioning errors with traditional elevation-dependent and refined stochastic models using the datasets from 7 stations on DOY 303–309, 2022 (the left panel displays the positioning errors in the first three hours at a converging stage, and the right panel displays those after a complete convergence).

integrations, namely B1C/B2b, B1C/B2b/B3I and B1C/B2b/B3I/B1I/B2a. As can be seen from the figure, the positioning performance of triple-frequency RT-PPP is better than that of dual-frequency one. The convergence time and the positioning accuracy of five-frequency RT-PPP have a significant improvement over the dual-frequency and triple-frequency RT-PPP.

Table 4 presents the mean values of the RMS errors and the convergence time at all stations during DOY 303–309, 2022. According to the statistical results, the kinematic RT-PPP positioning performance based on the B1C/B2b/B3I integration is slightly improved when compared with the B1C/B2b dual-frequency integration. By contrast, the kinematic RT-PPP with the five-frequency integration is significantly improved by 6 %, 32 %, and 9 % for the convergence time and by 11 %, 9 %, and 6 % for the positioning accuracy in east, north and up directions.

5.4. Performance of refined stochastic model

Using the formula in Chapter 4, the MP values of each frequency and the elevations of each satellite are calculated using the observations at seven stations shown in Fig. 1. Fig. 4 shows the scatterplot of code MP values for each frequency against the satellite elevation angle at URUM station on DOY 303, 2022. It can be seen from the figure that

Table 7
Five-frequency kinematic RT-PPP performance using traditional elevation-dependent and refined stochastic models.

Direction	Convergence time (minute)		Positioning accuracy (cm)	
	Traditional	Refined	Traditional	Refined
East	35.5	32.1	10.1	9.9
North	7.8	6.7	6.2	6.5
Up	21.0	19.1	14.3	14.0

the relationship between MP values and elevation angles of each frequency is in line with the base law that the MP decreases with the increase of elevation angles. In terms of the noise level, the code noise on the B2b and B2a frequencies is obviously smaller than that of the other three frequencies. By contrast, the code noise for the B1I frequency is the largest.

Table 5 provides the mean values of the MP RMS at each frequency for all stations. From this table, the MP values of the code observations at different frequencies of BDS-3 have obvious differences. Specifically, the code quality of the new frequency signals B2b and B2a is better, followed by the B1C and B3I signals, and the code quality of B1I is the worst. Additionally, the code MP values of the same frequency with different equipment (receiver and antenna) are different, while the code MP values of the same frequency with the same equipment are also different. So, we can conclude that the MP values may depend on the configurations of the station. However, we don't care about the exact value of MP value, but rather the ratio of MP values among different frequencies. Therefore, when establishing the stochastic model for multi-frequency RT-PPP, the code observations of different frequencies cannot be simply treated as equal weights. Instead, the MP values of different frequencies are used to modify the elevation-dependent stochastic model.

Using the carrier phase quartic difference method, the carrier phase noises at each frequency at each station are calculated. Fig. 5 shows the scatter chart of phase noises at each frequency at URUM station. It can be seen from the figure that the phase noises for the B2b and B2a frequencies are the smallest, while the noise level for the other three frequencies is similar.

The average RMSs of the carrier phase noises at each frequency of all stations are shown in Table 6. As can be seen from Table 6, there is no significant difference in the noise levels for carrier phase observations at different frequencies of BDS-3. Specifically, the carrier phase quality for the new frequencies of B2b and B2a is better, followed by the B1C and B1I frequencies, and the carrier phase quality of the B3I frequency is the worst. Actually, there should be no difference among frequencies as the phase observation (in cycles) is just the measurement of the carrier wave. However, there are still slight differences in phase noise (in meters) among different frequencies as shown in Table 6. This is mainly because the wavelength and the anti-interference ability of phase observations at different frequencies differ from each other.

By analyzing the noise of code and carrier phase observations at different frequencies, it can be concluded that the MP values of the code observations at different frequencies of BDS-3 have obvious differences. In particular, the new frequencies B2b and B2a have better code quality, followed by the B1C and B3I, and the code quality of B1I is the worst. In terms of phase noise, the noise level of the carrier phase has no significant difference at different frequencies. Therefore, based on the calculated code MP values and carrier phase noises above, the observation weight of each frequency relative to the B1I frequency can be determined using Eqs. (13) and (14).

To verify the effect of the refined stochastic model, the observations at the stations shown in Fig. 1 during DOY 303–309, 2022 are used to carry out kinematic B1C/B2b/B3I/B1I/B2a five-frequency UC RT-PPP with the traditional elevation-dependent stochastic model and the proposed refined stochastic model, respectively. Fig. 6 shows the epoch-wise RMS statistics of RT-PPP positioning errors with traditional elevation-dependent and refined stochastic models using the datasets from 7 stations on DOY 303–309, 2022. Following the figure, the proposed refined stochastic model shows a slightly faster convergence time.

Table 7 provides the mean values of the RMS errors and the convergence time for all stations during DOY 303–309. According to the statistical results, the proposed refined stochastic model performs better than the traditional elevation-dependent stochastic model in terms of convergence time with an improvement of 10 %, 14 % and 9 % in the east, north and up directions, respectively. Regarding the positioning accuracy, no obvious accuracy improvement is found.

6. Conclusions

BDS-3 satellites can broadcast real-time corrections of the precise satellite orbit, satellite clock, and DCB through PPP-B2b signals. Using these corrections, the RT-PPP can be achieved. This paper investigates the performance of the kinematic UC RT-PPP based on PPP-B2b from the perspective of different multi-frequency integrations, and stochastic model refinement.

The RT-PPP performance based on the B1I/B3I and B1C/B2b integrations is compared, and results indicate that the performance of the BDS-3 B1C/B2b integration is better than that of the B1I/B3I integration. The convergence time in the east, north, and up directions are shortened by 16 %, 15 %, and 12 %, respectively, and the positioning accuracies have a slight improvement in three directions. This paper also compares the positioning performance using the B1C/B2b dual-frequency integration, B1C/B2b/B3I triple-frequency integration and B1C/B2b/B3I/B1I/B2a five-frequency integration. The experimental results show that the kinematic PPP performance with the B1C/B2b/B3I integration is slightly improved when compared with the B1C/B2b dual-frequency integration, and the B1C/B2b/B3I/B1I/B2a five-frequency integration achieves a significant improvement by 6 %, 32 %, and 9 % in convergence time and by 11 %, 9 %, and 6 % in positioning accuracy in three directions, respectively.

The code MP and carrier phase noise are evaluated. Analysis results show that the noise of the code observations at different frequencies is quite different. The average MP values of the signals of B1I, B3I, B1C, B2b, and B2a are 0.216, 0.136, 0.139, 0.114, and 0.107 m, respectively, and the phase noise at each frequency has no significant difference. Based on this conclusion, a refined stochastic model is proposed by taking the MP and phase noise into account. Taking B1I as the reference frequency, different weights are assigned to the observations of different frequencies. The assigned weights are inversely proportional to the MP values and the phase noises. The experiment results indicate that the five-frequency kinematic UC RT-PPP with the refined stochastic model can improve the convergence time by 10 %, 14 %, and 9 % in east, north, and up directions when compared with the traditional elevation-dependent stochastic model, respectively. The convergence time and positioning accuracy of BDS-3 five-frequency kinematic PPP-B2b UC RT-PPP can reach 32, 7, and 19 min, and 9.9, 6.5, and 14.0 cm in the three directions, respectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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